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COS 202

Smart library circulation & Automation System

PROJECT OVERVIEW

A university library wants a **Smart Library Circulation & Automation System (SLCAS)** that manages books, users, borrowing/returning workflows, automated reminders, and interactive GUI dashboards.
You are required to build a fully functional system demonstrating advanced object-oriented design, efficient data structures, sorting/searching algorithms, recursion, and GUI programming with event-driven logic.

CORE REQUIREMENTS

1. Advanced Object-Oriented Programming (OOP)

Students must design a class hierarchy with:

- **Abstract class** LibraryItem
- **Subclasses:** Book, Magazine, Journal
- **Interface:** Borrowable
- **Polymorphism:**
    - A function that processes any LibraryItem type
- **Encapsulation & Composition:**
    - LibraryDatabase containing collections of items
    - UserAccount class with borrowing history
- **Class Organization:** Use multiple packages/modules

2. Class Hierarchies & Program Organisation

Project must be structured as:

```
/model
  LibraryItem.java
  Book.java
  ...
/controller
```

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LibraryManager.java

SearchEngine.java

BorrowController.java

/gui

MainWindow.java

BorrowPanel.java

AdminPanel.java

/utils

IDGenerator.java

FileHandler.java

3. Using APIs: Arrays, Lists, Queues, Stacks

System must use:

- **ArrayList** to store library items
- **Queue** for reservation/waitlist for borrowed books
- **Stack** for undoing last admin operation (e.g., deletion)
- **Array** to implement a quick fixed-size cache for "Most Frequently Accessed Items"

4. Search Algorithms (Student-implemented)

Implement at least **two**:

1. Linear search
2. Binary search
3. Recursive search (title or author)

Use the appropriate one depending on sorting state.

5. Sorting Algorithms (Student-implemented)

Implement at least **two**:

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1. Selection Sort
2. Insertion Sort
3. Merge Sort (recommended)
4. Quick Sort

Students choose which to use when sorting:

- by title
- by author
- by year

Include a GUI dropdown to pick algorithm.

6. Recursive Algorithm

At least one recursive component, e.g.:

1. Recursive search in the library catalogue
2. Recursively compute total resource count by category
3. Recursive menu refresh (console fallback)
4. Recursive overdue charge computation

7. Event-Driven Programming

GUI must respond to events such as:

- Button clicks
- Menu selections
- Text field updates
- Timer-triggered overdue reminders

Use listeners/handlers.

8. GUI Programming + Components + Layout Management

Students must create:

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- **Main window**
- **Tabbed panels**:
    - View Items
    - Borrow/Return
    - Admin
    - Search & Sort

GUI must include:

1. Tables or list views
2. Buttons (Add, Borrow, Return, Sort)
3. Text fields
4. Combo boxes
5. Labels
6. Dialog boxes
7. Status bar

Layouts:

1. BorderLayout
2. GridBagLayout or BoxLayout (encouraged)
3. CardLayout (for switching screens)

9. Advanced GUI Techniques

At least two of:

1.0Custom renderers for list/table items

2.0Dynamic components (adding/removing fields at runtime)

3.0File chooser dialog for import/export

4.0Timers for overdue notifications

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5.0Input validation with dialog popups

6.0Keyboard shortcuts / mnemonics

7.0Tooltips

10. GUI Application & Program Development

Final system must be:

1. Interactive
2. Robust
3. Error-tolerant (try/catch for invalid data)
4. Persistent (data saved & loaded using text/JSON files)

MANDATORY SYSTEM FEATURES

✔ **Add new library items**

✔ **Borrow and return items**

✔ **Maintain reservation queue**

✔ **Search items by title, author, or type**

✔ **Sort using selectable algorithms**

✔ **Undo last admin action (Stack)**

✔ **GUI dashboard with multiple panels**

✔ **Save & load data from files**

✔ **Generate simple reports:**

1. Most borrowed items
2. Users with overdue items
3. Category distribution

SUBMISSION REQUIREMENTS

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Students must submit:

- Source code folder
- Screenshots of GUI
- A diagram showing class hierarchy (UML encouraged)
- A 2–3 page report including:
    - Description
    - Features
    - Data structures used
    - Algorithms chosen and why
    - Challenges faced

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